HOMOGENEOUS COORDINATES: RATIOS, RATIONAL POINTS AND LIGHT

Triples and colour: keep the ratios

A triangle's shape and a light mixture's proportions both survive a change of scale.

1 / ONE POINT, MANY TRIPLES

Homogeneous coordinates remember proportions

The notation $(X : Y : Z)$ represents every non-zero multiple of a non-zero triple:

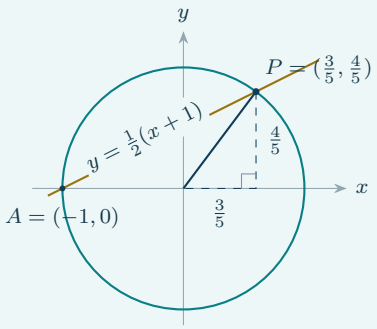
$$(X : Y : Z) = (\lambda X : \lambda Y : \lambda Z), \qquad \lambda \neq 0. \quad \text{For example, } (3 : 4 : 5) = (6 : 8 : 10).$$

When $Z \neq 0$, divide by Z to read the ordinary point $(X/Z, Y/Z)$. Thus both labels above give $(3/5, 4/5)$. Absolute scale has been discarded.

2 / A CIRCLE WRITTEN WITHOUT DENOMINATORS

Pythagorean triples lie on the conic $X^2 + Y^2 = Z^2$

Substitute $x = X/Z$ and $y = Y/Z$ into $x^2 + y^2 = 1$. Multiplying by Z^2 gives an equation whose terms all have degree 2: scaling X, Y, Z multiplies both sides by the same factor λ^2.



$(3 : 4 : 5)$ and $(6 : 8 : 10)$ describe the same point and similar right triangles.

A line $y = t(x + 1)$ through A meets the circle again at

$$x = \frac{1 - t^2}{1 + t^2}, \qquad y = \frac{2t}{1 + t^2}.$$

Put $t = n/m$ and clear denominators:

$$(m \ : \ n) \ \longmapsto \ (m^2 - n^2 \ : \ 2mn \ : \ m^2 + n^2).$$

Scaling (m, n) by $k \neq 0$ scales this triple by k^2.

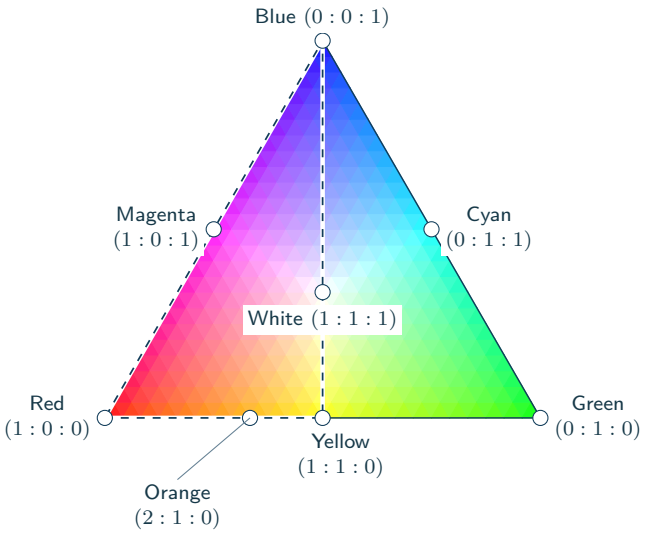
$(2 : 1) \mapsto (3 : 4 : 5), \quad (3 : 2) \mapsto (5 : 12 : 13), \quad (4 : 1) \mapsto (15 : 8 : 17).$

For a primitive positive triple, take coprime $m > n > 0$ of opposite parity. Every primitive triple occurs, allowing the legs to swap.

3 / A COLOUR TRIANGLE IS A SLICE THROUGH RAYS

Choose RGB light, then change to red–yellow–blue coordinates

For fixed, calibrated RGB primaries, let $r, g, b \geq 0$ be **linear-light intensities**. The ratio $(r : g : b)_{\mathrm{RGB}}$ records mixture proportions (chromaticity). Multiplying all three by $\lambda > 0$ changes intensity and preserves those proportions.



All diagram labels use RGB coordinates.
Dashed: the red–yellow–blue subtriangle.
Orange is the chosen $2 : 1$ red-to-green mix.

To draw the RGB triangle, choose the representative whose coordinates sum to 1:

$$(\rho, \gamma, \beta) = \frac{(r, g, b)}{r + g + b}.$$

Now use primary vectors $\mathbf{R} = (1, 0, 0)$, $\mathbf{Y} = (1, 1, 0)$ and $\mathbf{B} = (0, 0, 1)$. Then $a\mathbf{R} + b\mathbf{Y} + c\mathbf{B} = (a + b, b, c)$, so

$$(a : b : c)_{\mathrm{RYB}} \longleftrightarrow (a + b : b : c)_{\mathrm{RGB}}.$$

	RGB	RYB
Red	$(1 : 0 : 0)$	$(1 : 0 : 0)$
Yellow	$(1 : 1 : 0)$	$(0 : 1 : 0)$
Blue	$(0 : 0 : 1)$	$(0 : 0 : 1)$
Orange	$(2 : 1 : 0)$	$(1 : 1 : 0)$
Magenta	$(1 : 0 : 1)$	$(1 : 0 : 1)$
Cyan	$(0 : 1 : 1)$	$(-1 : 1 : 1)$

Cyan lies outside the RYB triangle: it needs $-\mathbf{R} + \mathbf{Y} + \mathbf{B}$. The negative coefficient describes a change of basis; positive RYB lights alone cannot make cyan.

Model. This is additive light. RYB paint mixing uses a different physical model. Screen RGB codes must be linearised before using them as light intensities. Black is $(0, 0, 0)$, which has no homogeneous point.

4 / THE SHARED IDEA AND ITS LIMIT

Choose a normalisation; remember what it removes

For the circle choose $Z = 1$; for colour choose $r + g + b = 1$. Both select one representative from each relevant ray. The first forgets a triangle's absolute size; the second forgets light intensity.

Actual light mixtures need intensities restored before addition: $(1, 0, 0) + (0, 0, 1)$ and $(2, 0, 0) + (0, 0, 1)$ give different proportions.

Read further: J. Lipman, *Rational Points on Conics*, p. 7; W3C, *CSS Color 4*, colour spaces and linear-light RGB.

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